

Targeted Behavioral Repair with Contaminated Anchors

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1 Abstract

We study the problem of correcting a specific, identified behavioral failure in a trained model without retraining from scratch and without full access to the original training distribution. Given a competent base model θ_0 and a small corrective dataset D_{fix} exhibiting the failure, the standard remedy for avoiding collateral damage elsewhere is to regularize the update toward θ_0 using an anchor set D_{anc} sampled from the original distribution. We highlight and formalize a complication that this standard recipe does not address: the anchor set is not guaranteed to be failure-free. When the failure region has nonzero mass under the original distribution, D_{anc} may itself contain instances of the very behavior being corrected, so a preservation term applied indiscriminately can directly oppose the corrective signal. We combine parameter-space preservation (Elastic Weight Consolidation) with output-space preservation (Knowledge Distillation) into a single objective, characterize the repair-preservation trade-off this creates as a function of failure prevalence ρ and dataset sizes, and analyze the resulting multi-step dynamics locally. The analysis shows that, under standard local assumptions, repeated updates converge to a curvature-weighted repair displacement determined by the preservation terms, while anchor contamination perturbs that equilibrium continuously, with the empirical estimate indicating a non-negligible contamination rate. We describe a concrete instantiation used to repair a behavioral-cloning policy trained on the StrikeForce environment.

2 Introduction

2.1 Problem

A model θ_0 that is already competent on its original task can still exhibit a specific, identifiable failure mode on a subset of the input space. Retraining from scratch is often infeasible: the original training set may no longer be fully available, or reacquiring it is costly relative to simply collecting examples of the failure itself.

2.2 Why Naive Fine-Tuning Fails

Fine-tuning θ_0 directly on failure examples D_{fix} corrects the targeted behavior but risks catastrophic forgetting [1]: nothing in the objective discourages the update from disrupting behavior the model already had right. The standard fix is to add a preservation term computed on an anchor set drawn from the original distribution, whether at the parameter level (EWC) or the output level (KD) [1, 2, 3].

2.3 The Contaminated-Anchor Complication

This standard recipe implicitly assumes the anchor set represents only desirable behavior. That assumption is not free: if the failure region $\mathcal{F}$ has nonzero probability $\rho = P_{\text{orig}}(x \in \mathcal{F})$ under the original distribution, an anchor set sampled from that same distribution will, in expectation, contain a ρ -fraction of failure examples. On those examples, the preservation term explicitly pulls the model back toward the very behavior the corrective term is trying to remove. We refer to this as the **contaminated-anchor** setting and treat it as a first-class part of the problem rather than an edge case to be assumed away.

2.4 Our Approach

We combine EWC (parameter-space preservation, weighted by Fisher-style sensitivity) and KD (output-space preservation through temperature-scaled softmax agreement with θ_0) into a single joint objective (Section 4). We then analyze the resulting multi-step repair dynamics and the effect of anchor contamination on the local preservation geometry (Section 6), followed by the practical repair-preservation trade-off in terms of ρ , $|D_{\text{fix}}|$, $|D_{\text{anc}}|$, and the regularization strengths λ_1, λ_2 (Section 7). We do not claim the contamination problem is solved outright; rather, we make its effect explicit so that the method’s behavior under it is predictable rather than silently absorbed into noise.

2.5 Summary of Contributions

- A formalization of the contaminated-anchor setting for behavioral repair, distinguishing it from standard continual-learning preservation, which typically assumes clean task boundaries.
- A combined EWC+KD objective (Section 4) with an explicit account of how the corrective and preservation gradients interact on contaminated anchor points (Section 5).
- A local multi-step analysis of the repair dynamics, showing that the joint update converges, under standard local assumptions, to a curvature-weighted repair displacement determined by the preservation geometry (Section 6).
- A characterization of how anchor contamination perturbs that preservation geometry and therefore the resulting repair point as a function of failure prevalence ρ (Section 6).
- A concrete implementation and repair pipeline applied to a behavioral-cloning policy for the StrikeForce environment (Section 8), with pre-repair (θ_0) and post-repair (θ^*) checkpoints evaluated on aggregate match metrics, a derived engagement ratio, and a raw-logit failure-rate metric (Section 11, outstanding).

3 Problem Setting

Let $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^K$ be a trained model with parameters θ . We assume an initial model θ_0 is already competent on the original task distribution P_{orig} , but exhibits a specific behavioral failure on a subset $\mathcal{F} \subseteq \mathcal{X}$. The goal is not to retrain from scratch, but to modify θ_0 so that the undesired behavior is corrected while preserving the model’s behavior outside $\mathcal{F}$.

We assume access to:

- a **corrective dataset** $D_{\text{fix}} = \{(x_i, y_i)\}_{i=1}^{n_f}$, containing examples of the undesired behavior with corrected labels;
- an **anchor dataset** $D_{\text{anc}} = \{x_j\}_{j=1}^{n_a}$, sampled from P_{orig} ;
- the original parameters θ_0 and the original model’s outputs on the anchor examples.

The original training set itself is assumed unavailable or too costly to reacquire, a setting also studied in continual-learning work such as Learning without Forgetting [3].

The failure condition is defined operationally rather than by a simple closed-form mathematical predicate. The concrete criterion used in this study is given in Section 11.2; $\mathcal{F}$ is used as a conceptual notation for the set of failure-inducing inputs or states, with concrete membership determined by that operational criterion.

Definition 1 (Contaminated-Anchor Assumption). *Let $\rho = P_{\text{orig}}(x \in \mathcal{F}) > 0$ denote the failure prevalence under the original distribution. Since D_{anc} is sampled from P_{orig} , ρ is equivalently the expected failure frequency within D_{anc} ; it is not the failure frequency within the corrective set D_{fix} , which is failure-dominated by construction: every frame in D_{fix} is an expert action recorded during an intervention.*

$$P_{\text{orig}} = (1 - \rho) P_{\text{good}} + \rho P_{\text{fail}},$$

and is therefore not assumed failure-free: $D_{\text{anc}} \cap \mathcal{F} \neq \emptyset$ in general.

This creates the central tension of the problem: a preservation mechanism applied indiscriminately to all anchor samples can oppose the desired repair.

4 Combined Objective

We seek $\theta^* \approx \theta_0$ in the sense of preserving general behavior, while improving performance on $\mathcal{F}$. The update $\Delta\theta = \theta^* - \theta_0$ should be targeted rather than globally disruptive.

4.1 Corrective Term

$$L_{\text{fix}}(\theta) = \frac{1}{n_f} \sum_{(x_i, y_i) \in D_{\text{fix}}} \ell(f_\theta(x_i), y_i).$$

D_{fix} is not intended to represent the full original distribution; it supplies a targeted signal for the specific missing or incorrect behavior.

4.2 Parameter-Space Preservation (EWC)

For each anchor example x_j , let $\ell_{\text{anc}}(x_j; \theta_0)$ denote the per-example loss used to characterize the original model’s behavior on that example. In the current implementation, the diagonal Fisher-style coefficient for parameter θ_i is estimated by accumulating the squared per-sample gradients and averaging over the anchor set:

$$\hat{F}_i = \frac{1}{n_a} \sum_{j=1}^{n_a} \left(\frac{\partial \ell_{\text{anc}}(x_j; \theta_0)}{\partial \theta_i} \right)^2.$$

Thus $\hat{F}_i$ is the mean squared sensitivity of parameter θ_i to the anchor-set loss. When the per-example loss is a negative log-likelihood under the usual Fisher construction, this is an empirical diagonal Fisher approximation; more generally, it should be interpreted as a squared-gradient sensitivity statistic. We use $\hat{F}_i$ in the EWC penalty

$$L_{\text{EWC}}(\theta) = \frac{1}{2} \sum_i \hat{F}_i (\theta_i - \theta_{0,i})^2.$$

Parameters with larger $\hat{F}_i$ therefore incur a larger cost for moving away from $\theta_{0,i}$ [1].

4.3 Output-Space Preservation (KD)

For anchor input x_j , let $z_0(x_j) = f_{\theta_0}(x_j)$ and $z_\theta(x_j) = f_\theta(x_j)$. With temperature T ,

$$p_0^T(x_j) = \text{softmax}\left(\frac{z_0(x_j)}{T}\right), \quad p_\theta^T(x_j) = \text{softmax}\left(\frac{z_\theta(x_j)}{T}\right),$$

and we use a forward-KL output-preservation objective,

$$L_{\text{KD}}(\theta) = \frac{1}{n_a} \sum_{j=1}^{n_a} \text{KL}(p_\theta^T(x_j) \| p_0^T(x_j)) = \frac{1}{n_a} \sum_{j=1}^{n_a} \sum_{k=1}^K p_{\theta,k}^T(x_j) \log \frac{p_{\theta,k}^T(x_j)}{p_{0,k}^T(x_j)}.$$

This is a KL-based behavioral anchoring mechanism related to the distillation literature [2, 4]; the present formulation specifically uses the positive forward KL implemented by the repair procedure.

4.4 Combined Objective

$$L_{\text{total}}(\theta) = L_{\text{fix}}(\theta) + \lambda_1 L_{\text{EWC}}(\theta) + \lambda_2 L_{\text{KD}}(\theta), \quad \lambda_1, \lambda_2 \geq 0.$$

5 The Conflict Induced by Anchor Contamination

If every anchor example represented desirable behavior, $\nabla_\theta L_{\text{EWC}}$ and $\nabla_\theta L_{\text{KD}}$ would generally align with the goal of behavioral preservation without opposing repair. Under Definition 1, this no longer holds unconditionally.

For $x \in D_{\text{anc}} \cap \mathcal{F}$, the reference output $f_{\theta_0}(x)$ is precisely the behavior the corrective term seeks to change. The KD gradient at such a point pulls $p_\theta^T(x)$ toward $p_0^T(x)$, i.e., back toward the failure behavior, directly opposing

$$\nabla_\theta L_{\text{fix}} \quad \text{vs.} \quad \nabla_\theta L_{\text{KD}}.$$

The method does not attempt to preserve θ_0 ’s behavior everywhere; it seeks an update in which the corrective signal is strong enough to overcome preservation pressure specifically on $\mathcal{F}$, while preservation remains effective on the (typically much larger) non-failure portion of the anchor set.

Remark 1. *This is an asymmetry, not a paradox: D_{fix} is concentrated on $\mathcal{F}$ by construction, while the contaminated fraction of D_{anc} is only $\rho \cdot n_a$ examples in expectation. The repair signal on $\mathcal{F}$ can dominate provided $|D_{\text{fix}}|$ and its effective weighting are large relative to $\lambda_2 \rho n_a$. Because the two gradient fields are architecture-, data-, and loss-dependent, we do not claim a closed-form sufficient condition for dominance; the present paper treats this dependence as a qualitative trade-off to be evaluated empirically rather than as a universal analytical guarantee.*

6 Local Multi-Step Repair Dynamics

The key theoretical question is not whether the corrective gradient dominates on the first optimization step, but where the sequence of updates moves after many steps. Define the preservation objective

$$R(\theta) = \lambda_1 L_{\text{EWC}}(\theta) + \lambda_2 L_{\text{KD}}(\theta),$$

so that

$$L_{\text{total}}(\theta) = L_{\text{fix}}(\theta) + R(\theta).$$

The repair procedure follows

$$\theta_{t+1} = \theta_t - \eta [\nabla L_{\text{fix}}(\theta_t) + \nabla R(\theta_t)].$$

6.1 Preservation Geometry Around the Original Model

Both preservation terms are minimized at θ_0 . For EWC this is immediate from the quadratic penalty. For the KL term, $p_\theta^T(x_j) = p_0^T(x_j)$ at $\theta = \theta_0$, so

$$R(\theta_0) = 0, \quad \nabla R(\theta_0) = 0.$$

Consequently, the first non-zero effect of preservation away from θ_0 is second-order. Let

$$H_R = \nabla^2 R(\theta_0).$$

A second-order expansion gives

$$R(\theta_0 + \Delta) = \frac{1}{2} \Delta^\top H_R \Delta + O(\|\Delta\|^3),$$

and

$$\nabla R(\theta_0 + \Delta) = H_R \Delta + O(\|\Delta\|^2).$$

For the EWC term,

$$H_{\text{EWC}} = \text{diag}(\hat{F}_1, \dots, \hat{F}_d).$$

Hence the implementation’s mean-squared per-sample gradient estimate enters the local preservation geometry directly as diagonal curvature.

For the KL term, let J_j be the Jacobian of the model logits with respect to θ at θ_0 for anchor x_j , and define

$$C_j^T = \frac{1}{T^2} \left[\text{Diag}(p_0^T(x_j)) - p_0^T(x_j) p_0^T(x_j)^\top \right].$$

At θ_0 , the Hessian of the KL term with respect to parameters is

$$H_{\text{KD}} = \frac{1}{n_a} \sum_{j=1}^{n_a} J_j^\top C_j^T J_j \succeq 0.$$

Therefore,

$$H_R = \lambda_1 \text{diag}(\widehat{F}) + \lambda_2 H_{\text{KD}} \succeq 0$$

locally. The EWC component penalizes parameter-space displacement according to squared gradient sensitivity, while the KD component penalizes displacement that changes the model outputs on the anchors.

6.2 Multi-Step Dynamics

Let

$$g_F = \nabla L_{\text{fix}}(\theta_0).$$

Within a sufficiently small neighborhood of θ_0 , linearizing the corrective loss and retaining the quadratic preservation term gives the local surrogate

$$L_{\text{local}}(\Delta) = L_{\text{fix}}(\theta_0) + g_F^\top \Delta + \frac{1}{2} \Delta^\top H_R \Delta, \quad \Delta = \theta - \theta_0.$$

The corresponding multi-step dynamics are

$$\Delta_{t+1} = (I - \eta H_R) \Delta_t - \eta g_F.$$

Thus the preservation force grows with displacement from θ_0 rather than appearing as a fixed competing vector. Early in the trajectory it is small; as the model moves farther away, the restoring term $H_R \Delta_t$ becomes stronger.

Proposition 1 (Local Equilibrium of the Repair Dynamics). *Assume that, on a subspace $\mathcal{S}$ containing the optimization trajectory, H_R is positive definite with eigenvalues in $[\mu, L]$, where $0 < \mu \leq L$, and suppose the local surrogate above is valid. If*

$$0 < \eta < \frac{2}{L},$$

then gradient descent on the local surrogate converges linearly to the unique stationary point

$$\Delta^* = -H_R^{-1} g_F$$

on $\mathcal{S}$. Moreover,

$$\|\Delta_t - \Delta^*\| \leq q^t \|\Delta_0 - \Delta^*\|, \quad q = \max_{\lambda \in [\mu, L]} |1 - \eta \lambda| < 1.$$

Proof. The stationary-point equation of the local surrogate is

$$g_F + H_R \Delta^* = 0,$$

which yields $\Delta^* = -H_R^{-1} g_F$. Subtracting the fixed-point equation from the update gives

$$\Delta_{t+1} - \Delta^* = (I - \eta H_R)(\Delta_t - \Delta^*).$$

Because the eigenvalues of H_R lie in $[\mu, L]$ and $0 < \eta < 2/L$, the spectral radius of $I - \eta H_R$ is strictly smaller than one, giving the stated linear convergence bound. $\square$

This result formalizes the multi-step interpretation of the method. The repair does not simply follow the corrective gradient indefinitely. Instead, the corrective signal drives the model away from θ_0 , while the preservation curvature increasingly resists displacement. Training settles when these forces balance:

$$\boxed{\nabla L_{\text{fix}}(\theta^*) + \nabla R(\theta^*) = 0.}$$

Under the local model, this balance is precisely $\Delta^* = -H_R^{-1}g_F$.

If H_R is only positive semidefinite, the same expression is understood on the identifiable/preserved subspace or with a small damping term γI replacing H_R by $H_R + \gamma I$. Without such a restriction, directions in $\ker(H_R)$ can permit movement that is invisible to the local preservation objective.

6.3 Interpretation as a Curvature-Weighted Repair

For a diagonal preservation geometry,

$$H_R = \text{diag}(h_1, \dots, h_d),$$

the equilibrium displacement satisfies

$$\Delta_i^* = -\frac{(g_F)_i}{h_i}.$$

Thus the method automatically permits larger movement in low-curvature directions and suppresses movement in high-curvature directions. The resulting repair direction is therefore not the raw gradient direction; it is the corrective gradient filtered through the local geometry of what the preservation terms consider important.

Under the local surrogate,

$$L_{\text{fix}}(\theta_0 + \Delta^*) - L_{\text{fix}}(\theta_0) \approx -g_F^\top H_R^{-1} g_F,$$

whereas

$$R(\theta_0 + \Delta^*) \approx \frac{1}{2} g_F^\top H_R^{-1} g_F.$$

Hence the same quadratic quantity

$$g_F^\top H_R^{-1} g_F$$

controls both the achievable first-order repair gain and the second-order preservation cost. This quantity is larger when the corrective gradient points through directions of low preservation curvature.

6.4 Behavior-Level Targetedness

Parameter displacement alone does not imply behavioral locality. To connect the parameter-space result to the desired behavioral property, let $z_G(\theta)$ and $z_F(\theta)$ denote the stacked model outputs on representative good and failure inputs, respectively, with Jacobians at θ_0

$$J_G = \left. \frac{\partial z_G}{\partial \theta} \right|_{\theta_0}, \quad J_F = \left. \frac{\partial z_F}{\partial \theta} \right|_{\theta_0}.$$

At the local equilibrium,

$$z_G(\theta^*) - z_G(\theta_0) \approx -J_G H_R^{-1} g_F,$$

while

$$z_F(\theta^*) - z_F(\theta_0) \approx -J_F H_R^{-1} g_F.$$

Therefore, a sufficient local condition for a targeted repair is that the preservation-weighted corrective direction has a substantially larger effect on the failure outputs than on the good outputs, for example

$$\boxed{\|J_F H_R^{-1} g_F\| \gg \|J_G H_R^{-1} g_F\|}.$$

A stronger structural version is the existence of a repair direction d such that

$$g_F^\top d < 0, \quad \|J_G d\| \approx 0, \quad \|J_F d\| > 0.$$

Such a direction improves the corrective objective while producing little first-order output change on good behavior. The EWC+KD geometry favors precisely those parameter directions that can produce the required repair with lower preservation cost. This does not constitute a global guarantee of behavioral equivalence; it is a local sufficient condition linking the parameter-space equilibrium to the desired behavior-level locality.

6.5 Effect of Anchor Contamination

Under the contaminated-anchor model,

$$P_{\text{anc}} = (1 - \rho)P_{\text{good}} + \rho P_{\text{fail}},$$

the population versions of both the squared-gradient sensitivity and KD curvature decompose accordingly:

$$H_R(\rho) = (1 - \rho)H_{R,\text{good}} + \rho H_{R,\text{fail}} + H_{\text{shared}},$$

where H_{shared} denotes any ρ -independent contribution such as a separately fixed preservation component. Equivalently, after absorbing the common term into $H_R(0)$,

$$H_R(\rho) = H_R(0) + \rho(H_{R,\text{fail}} - H_{R,\text{good}}).$$

Thus contamination changes the preservation geometry linearly in ρ at the population level.

Let

$$E_\rho = \rho(H_{R,\text{fail}} - H_{R,\text{good}}).$$

If $H_R(0) \succeq \mu I$ and

$$\|E_\rho\| < \mu,$$

then $H_R(\rho)$ remains positive definite and standard matrix perturbation gives

$$\|H_R(\rho)^{-1} - H_R(0)^{-1}\| \leq \frac{\|E_\rho\|}{\mu(\mu - \|E_\rho\|)}.$$

Consequently, with the same local corrective gradient g_F ,

$$\boxed{\|\Delta^*(\rho) - \Delta^*(0)\| \leq \frac{\|E_\rho\|}{\mu(\mu - \|E_\rho\|)} \|g_F\| = O(\rho)}$$

for sufficiently small ρ . Hence rare failure cases in the anchor set perturb the local repair equilibrium continuously rather than changing it discontinuously. The contamination remains harmful because its curvature points toward preserving the old failure behavior, but when ρ is small its aggregate influence is correspondingly limited under the stated conditioning assumptions.

With the empirically estimated $\hat{\rho} \approx 0.185$ (Section 8.3), the condition $\|E_\rho\| < \mu$ is correspondingly harder to satisfy, and the $O(\rho)$ bound should be interpreted as a first-order reference rather than a guarantee that the perturbation is negligible. In practice, this means the contaminated-anchor effect is expected to be a leading-order contribution to the repair equilibrium rather than a small correction, which is consistent with the framing of the problem as a first-class concern rather than an edge case.

6.6 Theoretical Interpretation

The resulting picture is therefore a multi-step balance rather than a one-step dominance argument:

$$\underbrace{\nabla L_{\text{fix}}}_{\text{repair drive}} + \underbrace{\lambda_1 \nabla L_{\text{EWC}} + \lambda_2 \nabla L_{\text{KD}}}_{\text{restoring preservation force}} = 0 \quad \text{at equilibrium.}$$

The EWC coefficients measured as mean squared per-sample sensitivities determine which parameter directions are expensive to change, while KD determines which anchor-output changes are expensive. When the corrective signal admits a low-preservation-cost direction, the equilibrium can produce a substantial correction with limited collateral movement. Anchor contamination perturbs this geometry in proportion to its prevalence, so the main failure mode of contamination is a shift of the equilibrium toward preserving part of the behavior that the repair objective seeks to eliminate.

The analysis is local and does not claim that neural-network optimization is globally convex, that the empirical Fisher equals the exact Fisher in every implementation, or that the resulting equilibrium necessarily gives perfect failure correction and zero general-performance loss.

7 Repair–Preservation Trade-off

As ρ decreases, most anchor samples represent desirable behavior and the preservation terms receive a strong aggregate signal from non-failure examples, reducing the conflict of Section 5. In the local analysis of Section 6, this corresponds to a smaller perturbation of the preservation Hessian $H_R(\rho)$. The corrective dataset provides a concentrated signal $D_{\text{fix}} \sim P(x \mid x \in \mathcal{F})$ regardless of ρ . The effective optimization is therefore governed jointly by $|D_{\text{fix}}|$, $|D_{\text{anc}}|\rho$, the preservation geometry, and the regularization strengths λ_1, λ_2 , rather than by either dataset size alone.

We do not assume a fixed value of λ_1, λ_2 is universally optimal; the method should be evaluated as a function of $|D_{\text{fix}}|, |D_{\text{anc}}|, \rho, \lambda_1, \lambda_2$.

Target Property

The desired solution satisfies, qualitatively,

$$\text{Failure}(\theta^*) < \text{Failure}(\theta_0), \quad \text{Performance}_{\text{good}}(\theta^*) \approx \text{Performance}_{\text{good}}(\theta_0).$$

The objective is a *localized* behavioral correction with limited collateral change, not merely a lower training loss.

8 Implementation

We instantiate the combined objective for a behavioral-cloning policy [9] trained in the *StrikeForce* 2D partially observable battle-arena testbed [10], using the decoupled Future Action Prediction (FAP)/Transformer-aggregator architecture described in [11]. We therefore apply the repair directly to the converged checkpoint θ_0 produced by that training procedure, without independently retraining the underlying policy architecture for the present study.

8.1 Precomputed Anchor Statistics

Before repair, we precompute and pin, for each anchor episode: θ_0 itself, an estimate of the Fisher diagonal $\{F_i\}$, and the reference future-prediction and policy logits $\{z_0(x_j)\}$ under θ_0 . Pinning the logits rather than recomputing $f_{\theta_0}(x_j)$ at every step avoids carrying a second live copy of the model during repair and fixes the KD reference exactly.

8.2 Per-Step Composition

At each optimization step we compute three gradient contributions and sum them before the optimizer step: (i) the corrective gradient from a shuffled minibatch of D_{fix} , using the same time-discounted focal loss as the original training procedure; (ii) the EWC gradient, evaluated over all parameters against the pinned (θ_0, F) ; (iii) the KD gradient, evaluated as a KL-type divergence between current and pinned softmax outputs at both the future-prediction heads and the main policy head.

Remark 2. *In the current implementation, the corrective term is minibatched (shuffled subsets of D_{fix} each step) while the KD term is evaluated over the entire anchor set at every step. This gives the anchor term zero gradient stochasticity relative to the corrective term and dominates per-step compute. The asymmetry is intentional: the anchor term is kept deterministic across optimization steps so that the preservation signal is applied consistently against the fixed reference outputs. This does not change the joint-objective gradient identity of Section 4, and full-batch KD is an implementation choice rather than a requirement of the method; minibatched KD would be a valid computational alternative.*

8.3 Anchor and Corrective Set Provenance

The corrective set D_{fix} is constructed from *intervention episodes* collected during unmasked inference. Concretely, while the policy is deployed in inference mode, it occasionally enters a state in which it produces an unpleasant, illogical, or otherwise unacceptable action. A human expert intervenes at that point, takes control, and executes the intended action until the policy has escaped the problematic situation, after which inference resumes. Each intervention episode therefore supplies a short trajectory of states paired with the expert’s actions, i.e. the corrected labels for the failure region. During repair, the model is trained only on these expert actions. The corrective set is thus concentrated on $\mathcal{F}$ by construction, and its failure density is close to one, in contrast to the anchor set.

The anchor set D_{anc} , by contrast, is sampled from the original distribution via the stage-1 validation episodes. It is not constructed to exclude failure cases: the behavioral failure is identifiable but is not assumed to be cleanly separable from the original distribution. Instead, the intended setting is that such cases are relatively rare, so most anchors still represent desirable behavior while a minority may overlap with $\mathcal{F}$. This is precisely the contamination scenario formalized in Definition 1.

Here ρ denotes the empirical failure frequency *within* D_{anc} , not within D_{fix} : the two sets have very different failure densities by construction. Every frame in D_{fix} is an expert action recorded during an intervention, so the failure density in D_{fix} is close to one. The anchor set, by contrast, contains only the ρ -fraction of frames that coincide with the failure region.

We estimate ρ empirically from the rate at which the policy required expert intervention during normal operation. Across 28 episodes of 1024 frames each (28,672 frames total), 5300 frames

required intervention, giving

$$\hat{\rho} = \frac{5300}{28672} \approx 0.185.$$

This estimate is non-negligible: the contaminated fraction of D_{anc} is a substantial minority rather than a vanishing edge case. The contaminated-anchor setting formalized in this paper is therefore a practical concern rather than a theoretical curiosity, and the conflict between the corrective and preservation gradients analyzed in Section 5 is expected to be observable in the experiments rather than masked by a vanishing ρ . When episode-level labels or behavioral inspection are available, the observed failure frequency in D_{anc} can be reported directly as an empirical estimate of the contamination rate; the method itself does not require explicit filtering of those examples.

8.4 Temperature

We use $T = 1$ in the current implementation, i.e., ordinary softmax rather than temperature-softened distillation targets.

9 Related Work

Elastic Weight Consolidation [1] penalizes movement of parameters deemed important to previous tasks, estimated via the Fisher information matrix; we use the same diagonal-Fisher penalty but apply it against a single identified failure region rather than a sequence of task boundaries. Knowledge distillation [2] was originally proposed for transferring information from a teacher to a student; here θ_0 and θ^* share an architecture, and KD instead serves as an output-space behavioral anchor. Learning without Forgetting [3] is also closely related to our data-availability constraint because it studies learning new capabilities when the original training data are unavailable, although its objective is preservation of existing capabilities while adding new ones rather than correcting a known failure.

More broadly, our setting belongs to the family of post-hoc model-editing approaches that seek to alter a pretrained model while limiting unintended changes. MEND [5] learns to transform gradients for fast, localized edits, ROME [6] directly modifies selected model parameters to rewrite factual associations, and task arithmetic [7] steers pretrained-model behavior through parameter-space task vectors. These approaches demonstrate the broader feasibility of targeted behavioral modification, but they address different editing mechanisms and problem formulations from the EWC+KD repair procedure studied here. In particular, our focus is not on editing factual associations in language models or learning a separate editor; it is on repairing a behavioral failure in an existing policy using a small corrective set plus limited anchors while explicitly accounting for anchor contamination.

The closest methodological precedent is [4], who combine a task-learning loss with EWC and KD to balance new-task learning against retention in a continual-classification setting. Our problem differs in framing: rather than sequential task learning, we address targeted correction of a single identified behavioral failure under incomplete access to the original training data, and we treat the possibility that the anchor set itself contains the failure ($D_{\text{anc}} \cap \mathcal{F} \neq \emptyset$) as a first-class part of the problem rather than an incidental detail. Recent analysis of model editing also shows that repeated edits can induce forgetting and downstream performance degradation [8], reinforcing the importance of evaluating collateral behavioral change rather than reporting only edit success.

10 Scope Statement

Explicitly not claimed:

- That D_{fix} or $\mathcal{F}$ admits a simple task-independent mathematical definition; failure is identified operationally from unmasked inference behavior, with task-specific or human judgment where needed.
- That preservation on D_{anc} guarantees global behavioral equivalence $f_{\theta^*}(x) \approx f_{\theta_0}(x)$ for all $x \in \mathcal{X}$; it is an empirical, regularized approximation over representative samples.
- That a specific (λ_1, λ_2) setting is universally optimal; the trade-off is a function of ρ , $|D_{\text{fix}}|$, and $|D_{\text{anc}}|$ and should be tuned per setting.
- That the local equilibrium analysis of Section 6 is a global convergence theorem for a non-convex neural network; its convergence result applies to the stated local surrogate and to the subspace where the preservation Hessian is positive definite.
- That the empirical diagonal Fisher estimate used here is the exact Fisher matrix in every loss construction; outside the standard likelihood setting, it is explicitly a squared-gradient sensitivity statistic.
- That the current implementation’s full-batch anchor evaluation is necessary to the method; it is a design choice of the present pipeline (Section 8).
- That the sequential learning setting of [4] and the single-failure-correction setting studied here are the same problem; we adopt their combined-loss mechanism but not their task-boundary framing.
- That the failure modes covered by the failure-rate evaluation exhaustively represent the space of failure-inducing states. It covers a representative subset (wall collision, adjacent enemy, flanking attacker); the full failure region is not cleanly enumerable.
- That the failure-rate metric is part of the default StrikeForce Lab Protocol. It is defined specifically for this repair study; the default protocol covers only aggregate match metrics.
- That the frames-per-kill ratio is a direct measurement of the engagement failure. It is a derived quantity computed from the aggregate benchmark, and serves as a proxy for how efficiently the policy converts survival into engagement; it does not isolate the specific failure modes analysed in Section 11.2.
- That the aggregate benchmark, the derived engagement ratio, and the raw-logit failure rate are computed from the same set of runs. The aggregate benchmark and the derived engagement ratio share one set of runs; the raw-logit failure rate is computed from an independent set of 10 inference runs, as described in Section 11.3. The three numbers are not derived from each other.
- That the qualitative inspection of the recorded runs is a substitute for quantitative failure-mode measurement. It is reported as illustrative evidence consistent with the aggregate and derived metrics, and is not used to derive numeric claims.

11 Experimental Design

11.1 Checkpoints Under Comparison

Three checkpoints are considered in the behavioral-repair study. We keep the original pre-repair and post-repair checkpoints exactly as obtained from the existing training and repair pipeline, and use naive fine-tuning as the corresponding ablation:

- θ_0 — **Pre-repair checkpoint:** the converged policy checkpoint before behavioral repair. This checkpoint is produced by the Future Action Prediction / Transformer-aggregator training procedure described in [11].
- θ_{FT} — **Naive fine-tuning baseline:** obtained by optimizing θ_0 on D_{fix} alone, i.e. setting $\lambda_1 = \lambda_2 = 0$. This baseline isolates the effect of applying the corrective signal without any explicit preservation mechanism.
- θ^* — **EWC+KD repair:** the checkpoint produced by the combined objective in Section 4, using the implementation in `ewc-kd.cpp`.

We apply the repair procedure directly to the converged checkpoint θ_0 produced by the training procedure described in [11]. Thus, the present experiments evaluate post-hoc behavioral repair on an already-trained policy rather than introducing or retraining a new policy architecture.

11.2 Failure-Case Identification

The targeted failure is identified from *unmasked inference*. During this inspection, the policy is evaluated without the deployment-time heuristic action mask so that the native behavior of the learned policy can be observed directly. This distinction matters because a downstream mask can suppress an undesirable action without changing the underlying policy tendency that the repair is intended to modify.

A failure case is a state in which the policy’s *raw* action preference is inconsistent with the intended task behavior. We identify failure states by inspecting the raw policy logits $f_{\theta_0}(s)$ before the inference-time action mask is applied. The mask suppresses specific undesirable actions at deployment time but does not alter the underlying policy preference: if the argmax of the raw logits still selects a failure action, the policy has not been repaired even when the masked output appears acceptable. Formally, a state s is a failure case if the task-correct action $a^*(s)$ — defined operationally as the action a competent human player would take in s , e.g., firing at an adjacent enemy, turning toward a flanking attacker, or avoiding a wall collision — is not the argmax of the raw logits $f_{\theta_0}(s)$. Evaluating on the raw argmax therefore measures whether the repair has actually altered the policy rather than whether the mask is doing the work.

This definition is operational and task-specific rather than a universal mathematical predicate. It covers a representative subset of the failure modes observed in the environment; the space of failure-inducing states is not cleanly enumerable, and the failure-rate evaluation introduced below is not claimed to exhaust it. We treat $\mathcal{F}$ as a conceptual notation for the failure region, with concrete membership determined by the operational criterion above.

Because the failure criterion is defined on the raw policy rather than the masked output, the same criterion can be applied to any checkpoint, and the resulting failure sets are directly comparable across θ_0 , θ_{FT} , and θ^* .

11.3 Evaluation Protocol

Evaluation uses the same inference setup as the aggregate match benchmark. All compared checkpoints use the same environment and the same inference-time post-processing rule so that differences between checkpoints are not confounded by an inconsistent deployment configuration.

- **Mask policy: uniform, always on.** The heuristic inference-time action mask (`bc_inference`) is applied uniformly to all compared checkpoints, matching the evaluation convention used for the reported benchmark.
- **Episode count.** Aggregate benchmark results for θ_0 , θ_{FT} , and θ^* are based on $N_{\text{episodes}} = 10$ episodes each.

A separate failure-rate evaluation reuses the same setup: each checkpoint is rerun for 10 inference runs, and the raw (unmasked) logits are recorded at every decision frame.

The two evaluations use independent sets of inference runs. The aggregate benchmark in Table 1 is computed from one set of 10 runs, and the raw-logit failure-rate in Table 2 is computed from a separate set of 10 runs. The two sets are disjoint: the failure-rate measurement does not reuse the trajectories underlying the aggregate metrics, and the resulting numbers are not derived from each other. This separation reflects the fact that the two evaluations use different post-processing conventions (the aggregate benchmark applies the inference-time action mask, while the failure-rate criterion is evaluated on the unmasked logits) and provides an independent check on the policy rather than a recomputation of the same trajectories.

11.4 Metrics

Match-level metrics are adopted directly from the StrikeForce Lab Protocol ¹ and reported per checkpoint:

$$T_{\text{alive}}, \quad N_{\text{kills}}, \quad A_{\text{HP}}, \quad A_{\text{stamina}}, \quad A_{\text{damage}}.$$

These measure preservation of general task performance.

The targeted repair objective is assessed separately on a failure-rate metric defined on the raw policy, following Section 11.5. At each decision frame we apply the operational criterion of Section 11.2 to the unmasked logits; the failure rate is the fraction of decisions on which the raw argmax does not match the task-correct action, aggregated over all runs. This metric is not part of the default Lab Protocol. The central empirical question is whether this rate decreases under θ^* relative to θ_0 while the aggregate metrics remain approximately preserved.

As a complementary indicator of the engagement failure — the policy’s tendency to survive without converting survival time into successful engagement — we additionally report the ratio of total decision frames to total kills for each checkpoint. This quantity is derived directly from the aggregate metrics in Table 1 and requires no additional runs. A larger value indicates that the policy spends more frames per successful engagement, which is characteristic of the engagement failure mode. Unlike the raw-logit failure rate, which is evaluated on a separate set of inference runs, this derived metric is computed from the same runs as the aggregate benchmark, and its sampling properties therefore coincide with those of Table 1.

¹<https://github.com/bistoyek21-ric/StrikeForce/blob/main/StrikeForce-client/StrikeForce-Lab-Protocol.md>

11.5 Failure-Rate Validation

To evaluate the repair directly on the failure region, we rerun each checkpoint under the same inference setup used for the aggregate match benchmark (Section 11.3), for 10 inference runs per checkpoint. To improve the reliability of the failure-rate estimate, we retain only runs containing at least 1024 decision actions. This minimum-length requirement matches the default episode length in the dataset and prevents unusually short runs from disproportionately affecting the measured rate. All 10 runs per checkpoint met this requirement; runs that did not were discarded at collection time and replaced before any failure-rate measurement, so the reported rates are aggregated over a full set of 10 valid runs per checkpoint. In addition, an episode terminates when the agent’s HP, stamina, or damage reaches its minimum value; these termination constraints prevent prolonged or cyclic behaviors from artificially inflating or deflating the observed failure frequency. These runs are distinct from the 28 fixed-length episodes used to construct D_{fix} in Section 8.3; the failure-rate runs involve no human intervention, and the raw logits are recorded at every decision frame.

At each decision frame s , we apply the operational failure criterion of Section 11.2 to the raw (unmasked) logits of the checkpoint under evaluation. A decision is counted as a *failure decision* if the argmax of the raw logits does not coincide with the task-correct action $a^*(s)$. The failure rate for a checkpoint is

$$\text{FailureRate}(\theta) = \frac{\#\{\text{failure decisions under } \theta\}}{\#\{\text{total decisions under } \theta\}},$$

aggregated over the retained inference runs.

Because the criterion is evaluated on the raw logits rather than the masked output, it measures the underlying policy tendency rather than the effect of the inference-time action mask. The mask suppresses the failure action at deployment time, so the failure does not necessarily manifest as an in-game action; the metric is defined on the policy’s raw preference so that it reflects whether the repair has actually altered the policy.

We report $\text{FailureRate}(\theta_0)$, $\text{FailureRate}(\theta_{\text{FT}})$, and $\text{FailureRate}(\theta^*)$ over the retained runs per checkpoint, with a per-failure-mode breakdown (e.g., wall collision, failure to engage an adjacent enemy, failure to turn toward a flanking attacker). The failure modes covered are a representative subset of those observed; the space of failure-inducing states is not cleanly enumerable, and this limitation is stated in the Scope Statement.

11.6 Results

Table 1 reports θ_0 , θ_{FT} , and θ^* , each evaluated over $N_{\text{episodes}} = 10$ episodes with the heuristic inference-time mask applied uniformly. Figures are reported as mean $\pm$ sample standard deviation across the 10 episodes.

Metric (mask always on)	θ_0 (pre-repair)	θ_{FT} (naive fine-tune)	θ^* (EWC+KD repair)
T_{alive}	11033.0 ± 5647.2	7057.0 ± 4103.4	14281.8 ± 5695.3
N_{kills}	133.5 ± 108.2	78.2 ± 47.9	215.0 ± 114.5
A_{HP}	$(1.158 \pm 0.560) \times 10^8$	$(0.844 \pm 0.450) \times 10^8$	$(1.523 \pm 0.627) \times 10^8$
A_{stamina}	$(0.964 \pm 0.517) \times 10^8$	$(0.729 \pm 0.343) \times 10^8$	$(1.212 \pm 0.387) \times 10^8$
A_{damage}	$(1.135 \pm 0.984) \times 10^7$	$(0.601 \pm 0.390) \times 10^7$	$(1.868 \pm 1.574) \times 10^7$

Table 1: General-performance comparison over $N_{\text{episodes}} = 10$ episodes each, with the heuristic inference-time mask applied uniformly. Values are mean $\pm$ sample standard deviation across the 10 episodes.

θ_{FT} performs below θ_0 on all reported metrics, while θ^* exceeds θ_0 on all five. Naive fine-tuning reduces mean survival from 11033.0 to 7057.0 and mean kills from 133.5 to 78.2; the EWC+KD repair increases these to 14281.8 and 215.0. The aggregate measurements should be read together with the per-episode variance and the failure-rate evaluation.

Table 2 reports the raw-policy failure rates measured independently for each checkpoint over 10 inference runs. The metric is evaluated on unmasked logits using the same operational failure criterion across all checkpoints, with runs retained only when they contain at least 1024 decision actions. Episodes terminate when HP, stamina, or damage reaches its minimum value.

Failure rate (raw logits)	θ_0 (pre-repair)	θ_{FT} (naive fine-tune)	θ^* (EWC+KD repair)
Overall	11.150%	13.754%	10.450%
Wall collision	4.810%	6.761%	4.110%
Adjacent enemy	1.422%	2.229%	2.499%
Flanking attacker	4.918%	4.764%	3.841%

Table 2: Failure rates computed from raw (unmasked) policy logits. Each checkpoint is evaluated independently over 10 inference runs using the same failure criterion and evaluation protocol. The reported values are aggregate rates over all decision frames in the corresponding runs.

Across the three checkpoints, the aggregate failure rates are 11.150% for θ_0 , 13.754% for θ_{FT} , and 10.450% for θ^* . The per-mode breakdown shows the same variation across checkpoints, with wall-collision, adjacent-enemy, and flanking-attacker failures contributing differently to the overall rate. These measurements are evaluated independently for each checkpoint under the same inference protocol and are therefore used as a direct empirical comparison of their raw policy behavior.

Table 3 reports a derived quantity computed from the aggregate metrics of Table 1: the total number of decision frames divided by the total number of kills. This ratio measures how many frames the policy spends, on average, before converting its survival into a successful engagement. A lower value indicates more efficient engagement; a higher value indicates that the policy survives without translating survival time into kills. Because it is computed directly from the totals in Table 1, it requires no additional runs and shares the sampling properties of the aggregate benchmark.

θ^* achieves the lowest ratio (66.4 frames per kill), compared to θ_0 (82.6) and θ_{FT} (90.2). This ordering is consistent with the qualitative inspection of the recorded runs described below and with the improvement in mean kills and survival time reported in Table 1. The derived ratio does not isolate the specific failure modes measured in Table 2; it is a coarse indicator of engagement efficiency, computed on a separate set of runs from the failure-rate measurement.

Metric	θ_0 (pre-repair)	θ_{FT} (naive fine-tune)	θ^* (EWC+KD repair)
Total kills	1335	782	2150
Total frames	110,330	70,570	142,818
Frames per kill	82.6	90.2	66.4

Table 3: Derived engagement metric computed from the aggregate benchmark of Table 1. Frames per kill is the ratio of total decision frames to total kills, aggregated over the 10 runs. A lower value indicates more efficient conversion of survival time into engagement.

Qualitative inspection of the recorded runs is consistent with the derived metric. In the θ_0 runs, the policy frequently fails to close distance to visible enemies even when no other tactical factor (low HP, unfavorable position) justifies retreat; the agent survives for extended periods without

converting its survival time into engagement. This behaviour is visible across the recorded runs and is reproducible by replaying the corresponding checkpoints in the StrikeForce environment. The θ^* runs, by contrast, show the agent actively pursuing visible enemies. These observations are qualitative and are reported alongside, not instead of, the quantitative metrics in Tables 1, 3, and 2; they serve to illustrate the behavioural difference that the derived engagement metric captures only indirectly.

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